

SEOUL SINYONGSAN, Korea, Republic of

WMO: 471170

Lat: 37.533N

Lon: 126.967E

Elev: 29

StdP: 100.98

Time Zone: 9.00 (E09)

Period: 82-05

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.8	-10.1	-20.0	0.6	-9.1	-18.0	0.8	-7.7	7.1	2.4	6.2	0.7	1.4	330	0.381

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.9	33.2	25.2	32.0	24.5	30.8	24.1	26.3	31.0	25.8	30.1	25.2	29.0	2.3	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.1	20.3	28.3	24.7	19.8	28.0	24.0	19.0	27.5	82.2	30.9	79.6	30.2	77.3	28.8	28.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.3	5.2	4.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.3	-2.5	0.5	6.2	13.7	18.3	23.1	26.1	26.5	21.8	15.7	8.3	1.5
	DBStd	10.45	4.05	4.58	3.49	3.88	2.73	2.51	2.55	2.38	2.98	3.55	4.14	4.38
	HDD10.0	1133	386	266	124	11	1	0	0	0	0	5	79	263
	HDD18.3	2620	644	499	376	144	34	2	0	0	5	93	302	521
	CDD10.0	2351	0	0	6	123	257	392	499	513	353	182	27	0
	CDD18.3	796	0	0	0	7	32	143	241	254	108	11	0	0
	CDH23.3	6193	0	0	0	45	186	912	2086	2338	593	33	0	0
	CDH26.7	2000	0	0	0	5	24	254	747	834	134	1	0	0
Wind	WSAvg	1.7	1.5	1.7	2.1	2.1	1.9	1.7	1.8	1.7	1.5	1.3	1.6	1.3
Precipitation	PrecAvg	1393	19	26	44	83	98	135	396	314	153	51	50	22
	PrecMax	2356	63	108	124	339	291	497	1131	1238	672	214	165	71
	PrecMin	761	1	0	3	8	8	16	115	67	5	0	8	3
	PrecStd	354	16	22	29	65	63	103	208	221	138	41	34	15
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.8	12.9	18.0	27.0	29.0	33.0	35.2	35.0	32.0	25.8	19.0	13.1
		MCWB	4.3	7.6	9.6	16.3	17.8	21.8	26.5	25.9	24.0	17.3	13.5	8.5
	2%	DB	6.0	10.2	15.8	24.2	27.0	31.0	33.8	33.2	29.9	24.1	17.1	10.2
		MCWB	3.1	6.1	8.6	15.9	17.3	21.4	26.0	25.4	22.8	17.1	12.5	6.9
	5%	DB	4.8	8.4	13.9	22.1	25.2	29.3	32.0	32.0	28.0	22.6	15.8	9.1
		MCWB	2.3	4.5	7.8	14.2	16.3	20.8	25.2	24.8	21.5	16.1	11.3	6.0
	10%	DB	3.2	6.8	11.9	20.2	24.0	28.0	30.2	30.9	26.2	21.0	14.1	7.9
		MCWB	0.9	3.3	7.0	13.2	16.1	20.3	24.4	24.7	20.5	15.5	10.4	4.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.0	9.0	11.1	19.6	20.8	24.1	27.4	27.1	25.4	19.3	15.2	9.9
		MCDB	7.0	11.1	16.0	23.4	24.7	29.0	32.8	31.6	29.1	22.8	17.7	12.1
	2%	WB	3.6	6.6	9.6	16.9	19.4	23.0	26.6	26.3	24.3	18.2	13.7	8.0
		MCDB	5.5	9.6	14.0	23.0	23.7	28.1	31.8	30.8	27.7	21.9	16.2	9.8
	5%	WB	2.5	4.7	8.4	15.3	18.2	22.1	25.9	25.9	23.4	17.3	12.1	6.2
		MCDB	4.3	8.1	12.9	21.1	22.6	27.1	30.4	30.2	26.5	20.9	14.7	8.8
	10%	WB	1.1	3.3	7.2	13.9	17.2	21.3	25.3	25.3	22.2	16.3	10.7	4.9
		MCDB	2.9	6.3	11.2	19.0	21.9	25.8	29.2	29.3	25.3	20.2	13.6	7.3
Mean Daily Temperature Range	5% DB	MDBR	7.5	7.9	8.7	10.4	9.5	8.6	6.7	6.9	8.2	9.1	8.2	7.4
		MCDBR	7.7	9.0	11.2	12.4	11.7	10.7	8.7	8.5	9.2	10.5	9.3	8.1
		MCWBR	6.0	6.1	6.5	6.2	5.2	4.5	3.5	3.4	4.2	5.1	5.8	5.9
	5% WB	MCDBR	7.4	8.2	10.4	11.0	9.5	8.8	7.3	7.2	7.7	9.0	8.1	7.3
MCWBR		6.3	5.9	6.7	6.2	5.2	4.3	3.2	3.3	4.2	5.2	6.1	6.1	
Clear-Sky Solar Irradiance	taub	0.441	0.496	0.579	0.626	0.638	0.710	0.678	0.589	0.516	0.509	0.489	0.439	
	taud	1.934	1.839	1.700	1.650	1.674	1.597	1.711	1.913	2.036	1.934	1.925	1.963	
	Ebn at Noon	690	704	686	687	689	640	657	708	737	685	636	658	
	Edh at Noon	141	176	222	246	244	264	234	187	157	158	142	129	
All-Sky Solar Radiation	RadAvg	2.50	3.27	4.27	4.97	5.59	5.23	4.07	4.34	4.13	3.58	2.44	2.19	
	RadStd	0.16	0.26	0.42	0.35	0.52	0.54	0.62	0.61	0.45	0.25	0.24	0.15	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+94	+50

Nomenclature: See separate page